Solution

Exercises: Continuous-Time Convolution

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1. Graphical evaluation of continuous-time convolution is used to find the (output, input) for a system given the (output, input) and the impulse response.

2. Why do we use graphical evaluation of convolution? Select all that apply.

a) Identify good numbers for playing the lottery

b) Develop insight regarding system behavior

c) Obtain an exact analytical expression for the system output

d) Prove that we can do integrals

3. Which of the following steps are involved in graphical evaluation of discrete-time convolution $y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau$. Select all that apply.

a) Find $x(t-\tau)$ as a function of τ for some fixed value t

b) Find $x(t-\tau)h(\tau)$ as a function of t for some fixed value τ

c) Integrate all the values in $x(t-\tau)h(\tau)$ over τ for some fixed value t

d) Find $x(t-\tau)$ as a function of t for some fixed value τ

e) Find $x(t-\tau)h(\tau)$ as a function of τ for some fixed value t

f) Integrate all the values in $x(t-\tau)h(\tau)$ over t for some fixed value τ

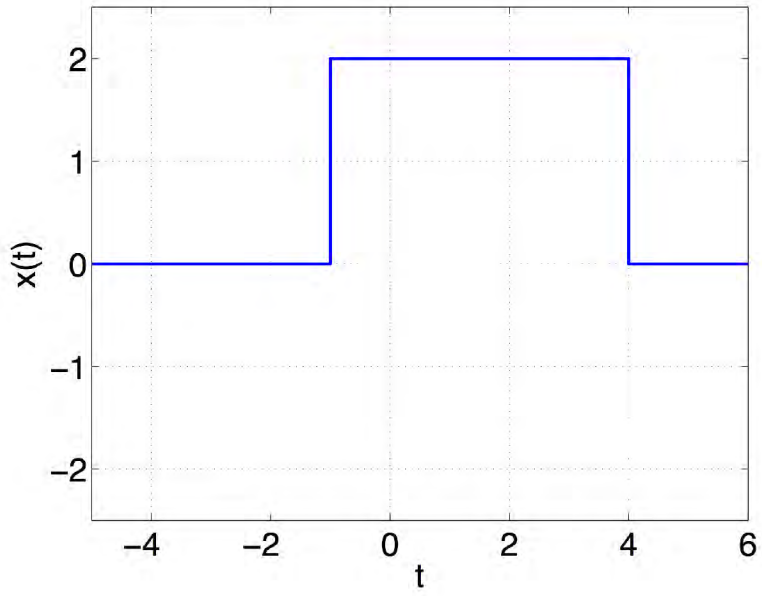
4. In order to find $x(t-\tau)$ for $t > 0$ you:

a) Reflect $x(\tau)$ about $\tau = 0$ to obtain $x(-\tau)$, then shift $x(-\tau)$ to the right for t seconds

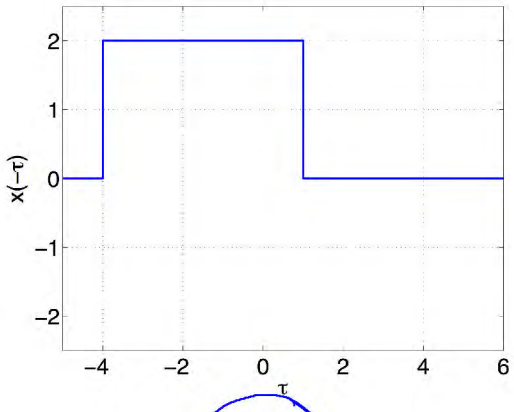
b) Reflect $x(\tau)$ about $\tau = 0$ to obtain $x(-\tau)$, then shift $x(-\tau)$ to the left for t seconds

c) Shift $x(\tau)$ to the right by t seconds, then reflect about $\tau = 0$

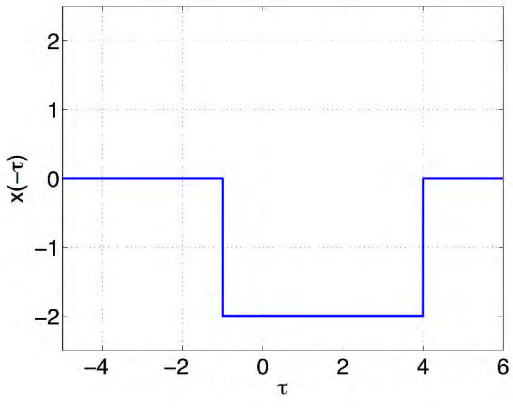
The continuous-time signal $x(t)$ is depicted in the figure below will be used in the next three problems. The signal is expressed mathematically as $x(t) = 2(u(t+1) - u(t-4))$.



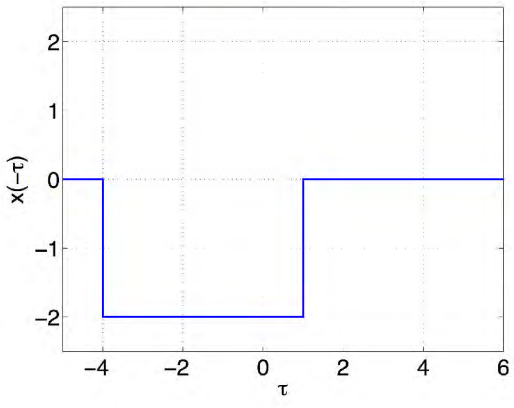
5. Select the answer corresponding to $x(-\tau)$.



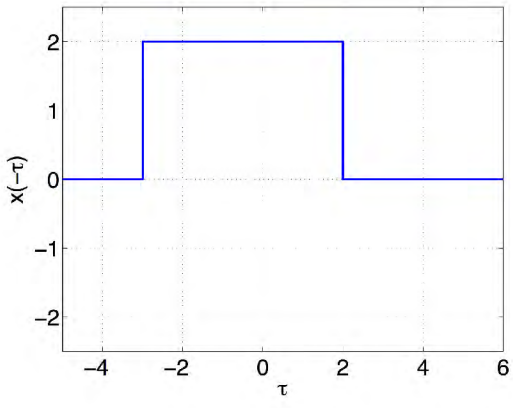
(a)



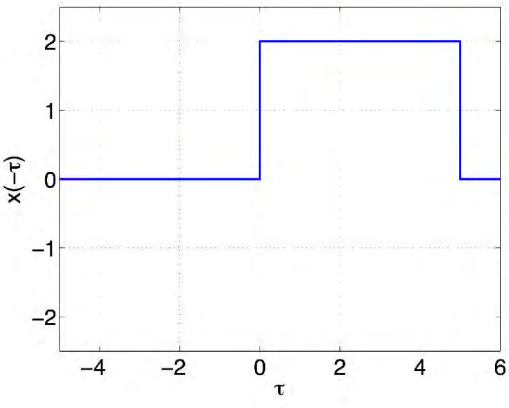
(b)



(c)

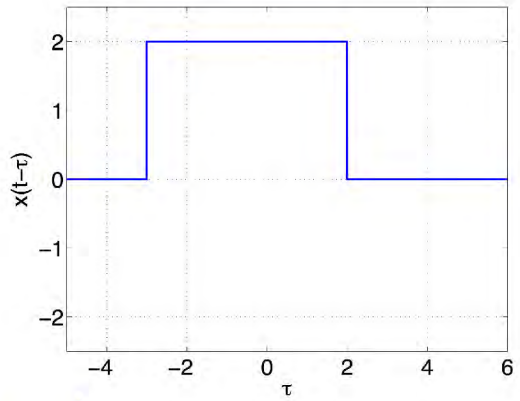


(d)

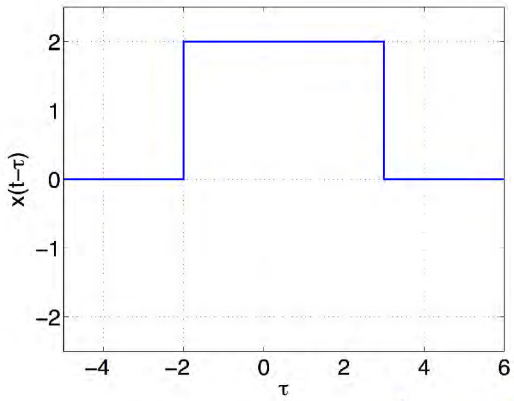


(e)

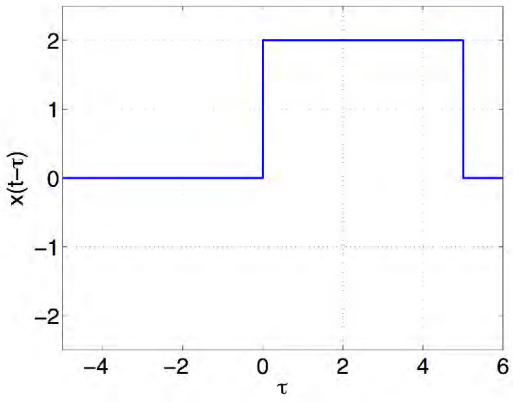
6. Each of the following graphs depict $x(t-\tau)$ for some value of t. Enter the t corresponding to each graph. You may assume t is an integer for convenience.



(g) $t=1$

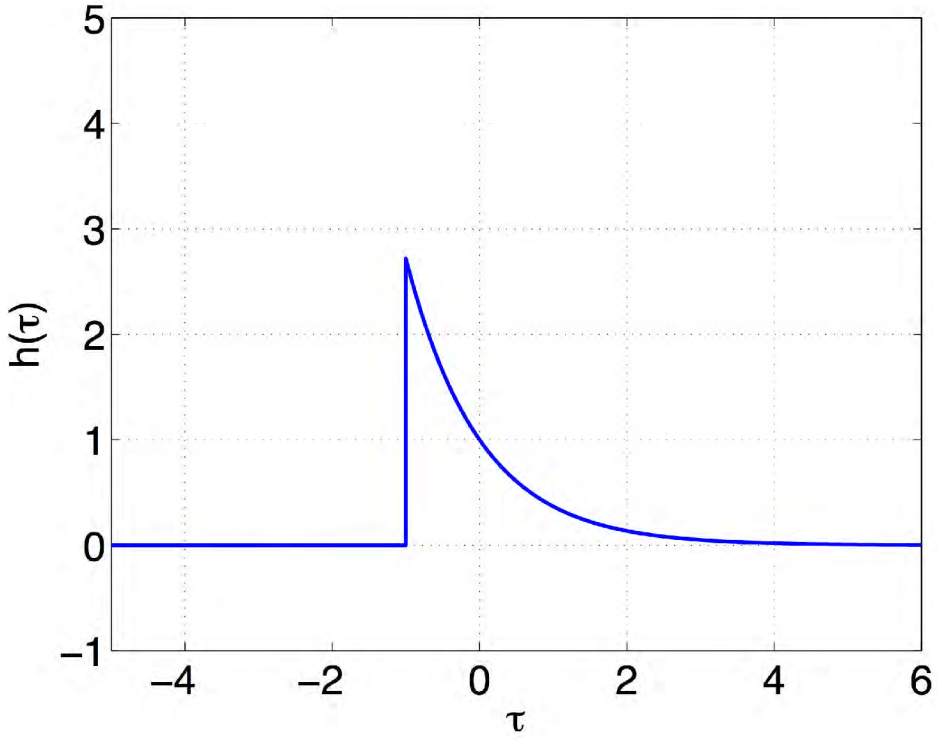


(h) $t=2$

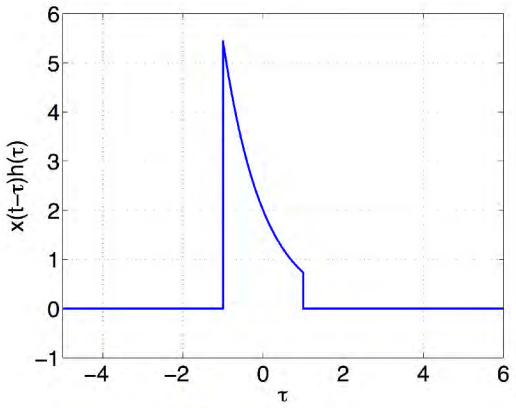


(i) $t=4$

7. Find the output of an LTI system using $y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau$ for the input $x(t)$ given previously. The impulse response is $h(t) = e^{-t}u(t+1)$ as shown below.

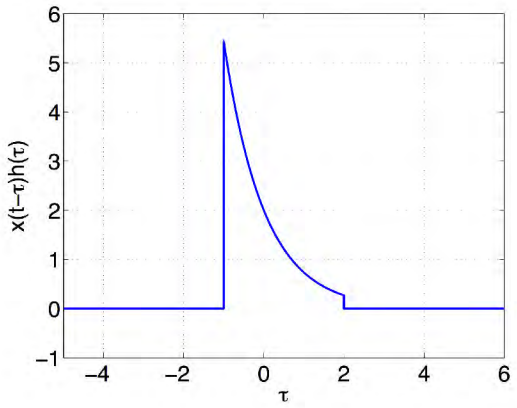


Each of the following graphs depict $x(t-\tau)h(\tau)$ for some value of t. Find the t corresponding to each graph. You may assume t is an integer for convenience.



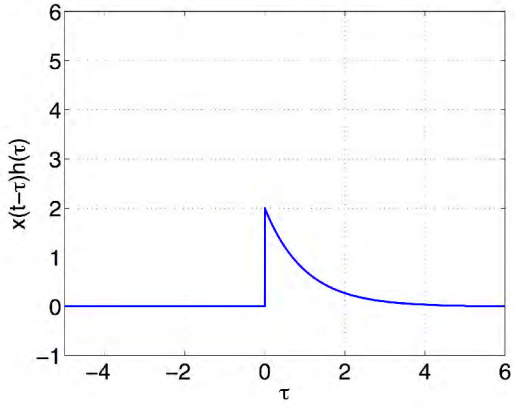
(j)

$t = 0$



(k)

$t = 1$

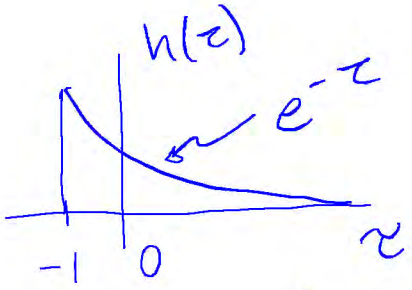


(l)

$t = 4$

Now find the output $y(t)$ for the following values of t. Hint:

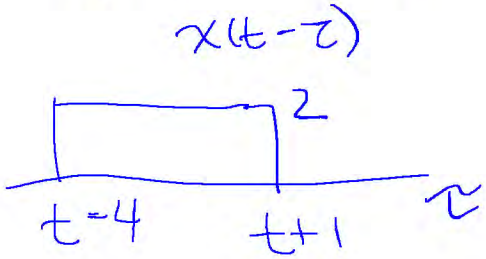
$$\int_a^b e^{-\tau} d\tau = e^{-a} - e^{-b}$$



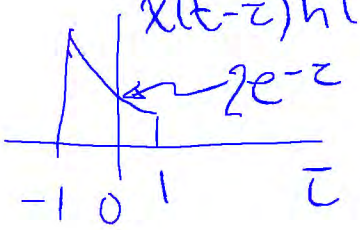
i) $y(-3.5)$

$x(t-\tau)h(\tau) = 0$

$y(t) = 0$



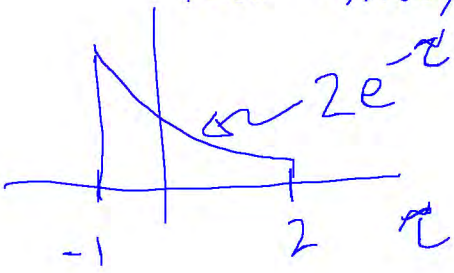
ii) $y(0)$



$$y(0) = 2\int_{-1}^{1} e^{-\tau} d\tau$$

$$= 2(e^1 - e^{-1})$$

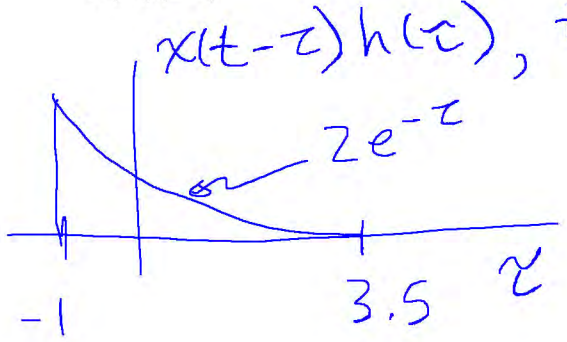
iii) $y(1)$



$$y(1) = 2\int_{-1}^{2} e^{-\tau} d\tau$$

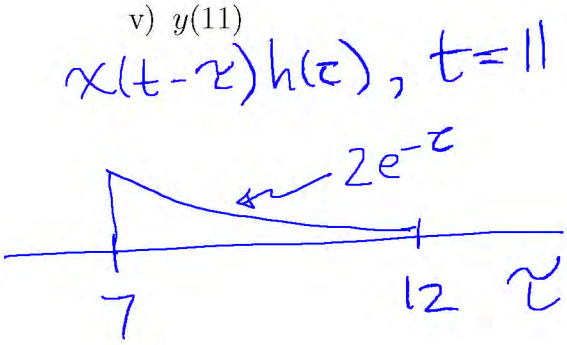
$$= 2(e^{+1} - e^{-2})$$

iv) $y(2.5)$



$$y(2.5) = \int_{-1}^{3.5} 2e^{-\tau}\,d\tau = 2\left(e^{1} - e^{-3.5}\right)$$

v) $y(11)$

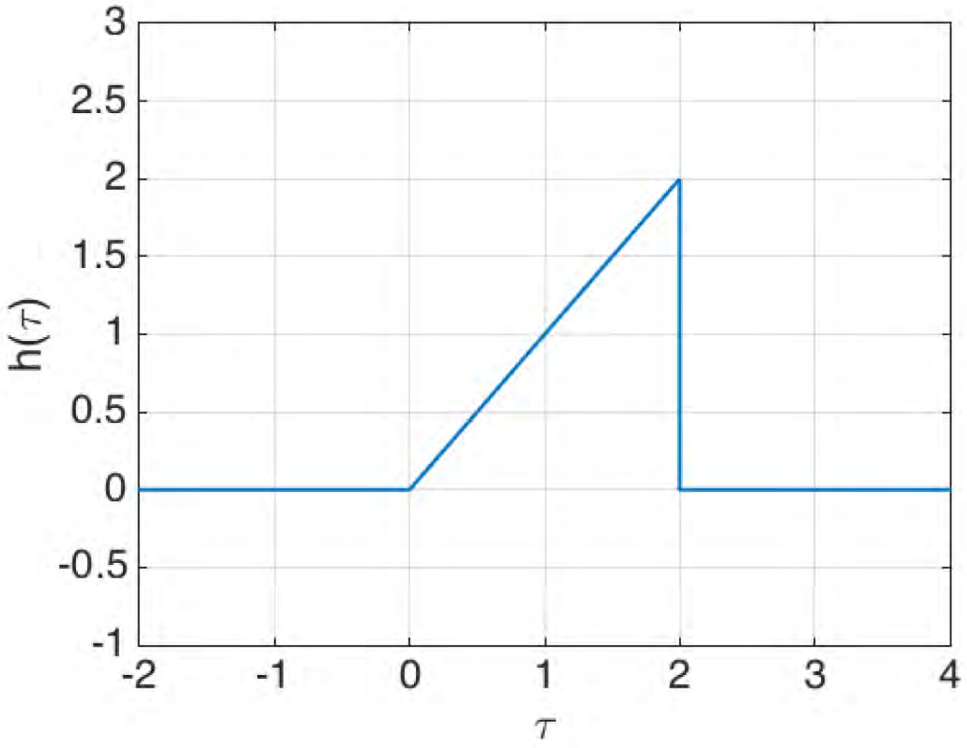


$$y(11) = 2\int_{7}^{12} e^{-\tau}\,d\tau = 2\left(e^{-7} - e^{-12}\right)$$

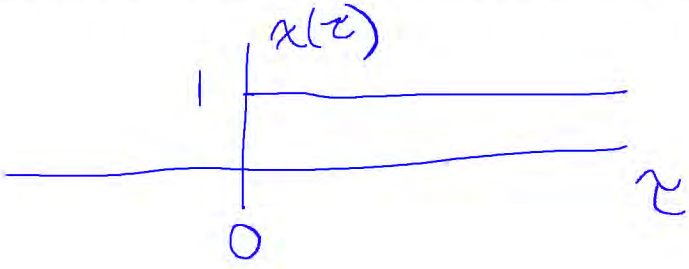
In general,

$$y(t) = \begin{cases} 0, & t < -2 \\ 2\left(e^{1} - e^{-(t+1)}\right), & -2 < t < 3 \\ 2\left(e^{-(t-4)} - e^{-(t+1)}\right), & t > 3 \end{cases}$$

8. In this problem we compute $y(t) = x(t) * h(t)]$ where $h(t) = t(u(t) - u(t-2))$ as shown below and $x(t) = u(t)$.



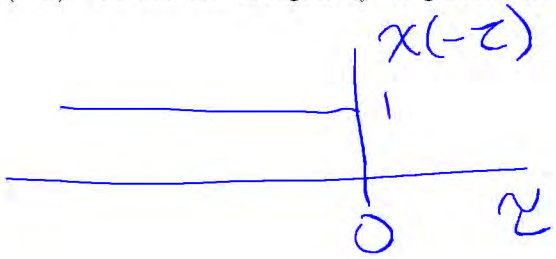
a) Graph $x(\tau)$. Enter the value of $x(\tau)$ on the following intervals.



i) $x(\tau) =$ 0 , $\tau < 0$

ii) $x(\tau) =$ 1 , $\tau > 0$

b) Graph $x(-\tau)$. Select the inequality in parentheses and enter the ranges of τ for



which $x(-\tau)$ takes the following values:

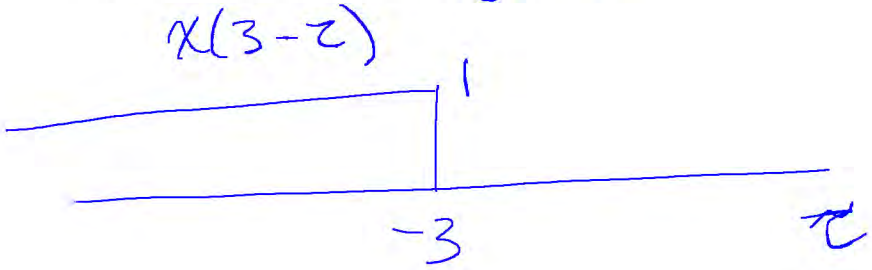
i) $x(-\tau) = 0$ for τ (> or <) 0

ii) $x(-\tau) = 1$ for τ (> or ≤) 0

c) Graph $x(t-\tau)$ for $t = -3$. Select the inequality in parentheses and enter the ranges of τ for which $x(t-\tau)$ takes the following values:

i) $x(t-\tau) = 0$ for τ (> or <) −3

ii) $x(t-\tau) = 1$ for τ (> or <) −3



d) Graph $x(t-\tau)$ for $t = 0$. Select the inequality in parentheses and enter the ranges of τ for which $x(t-\tau)$ takes the following values:

i) $x(t-\tau) = 0$ for τ (> or <) 0

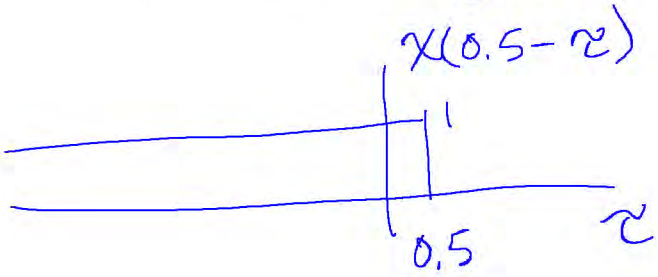
ii) $x(t-\tau) = 1$ for τ (> or ≤) 0

x(-τ) when t = 0 ⇒ see part b

e) Graph $x(t-\tau)$ for $t = 0.5$. Select the inequality in parentheses and enter the ranges of τ for which $x(t-\tau)$ takes the following values:

i) $x(t-\tau) = 0$ for τ (> or <) 0.5

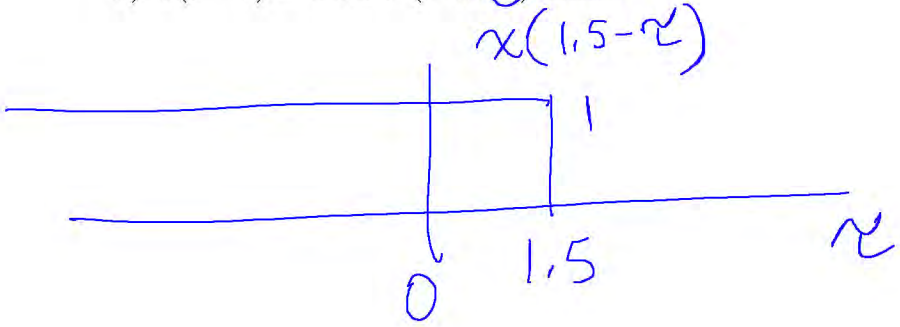
ii) $x(t-\tau) = 1$ for τ (> or ≤) 0.5



f) Graph $x(t-\tau)$ for $t = 1.5$. Select the inequality in parentheses and enter the ranges of τ for which $x(t-\tau)$ takes the following values:

i) $x(t-\tau) = 0$ for τ (> or <) 1.5

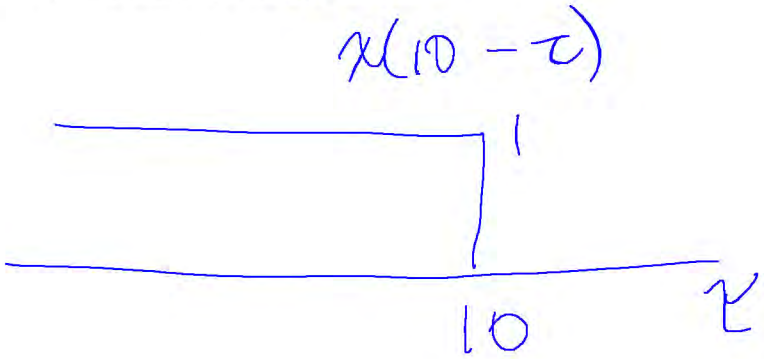
ii) $x(t-\tau) = 1$ for τ (> or ≤) 1.5



g) Graph $x(t-\tau)$ for $t = 10$. Select the inequality in parentheses and enter the ranges of τ for which $x(t-\tau)$ takes the following values:

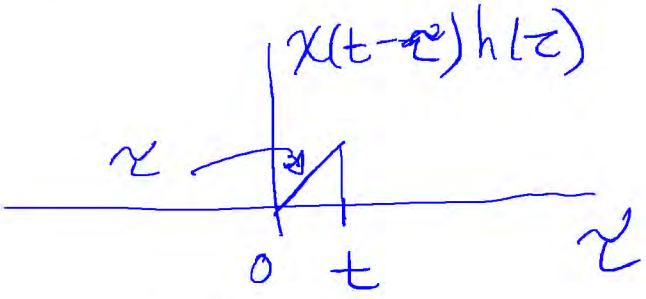
i) $x(t-\tau) = 0$ for τ (> or <) 10

ii) $x(t-\tau) = 1$ for τ (> or ≤) 10

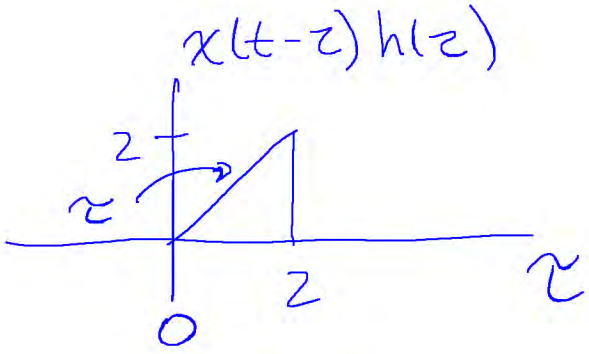


h) Graph $x(t-\tau)h(\tau)$ for several representative values of t.

$x(t-\tau)h(\tau) = 0$ for $t < 0$



$0 < t < 2$



$t > 2$

i) For what values of t is $x(t-\tau)h(\tau) = 0$? For t ($>$ or $<$) 0

ii) For what values of t is $x(t-\tau)h(\tau) \neq 0$ and unchanging as t increases? For t ($>$ or $<$) 2

iii) For what values of t does $x(t-\tau)h(\tau) = \tau(u(\tau) - u(\tau - t))$?

$0 < t < 2$

i) Now find $y(t)$ for the following values of t. Hint:

$$\int \tau d\tau = \frac{t^2}{2} + C$$

i) $y(-2.5)$

$y(t) = 0$ for $t < 0$

ii) $y(0.5)$ $y(0.5) = \int_0^{0.5} x\,dx = \frac{(0.5)^2}{2}$

$= 0.125$

iii) $y(1)$ $y(1) = \int_0^1 x\,dx = \frac{1^2}{2} = 1/2$

iv) $y(2)$ $y(2) = \int_0^2 x\,dx = \frac{2^2}{2} = 2$

v) $y(10)$ $y(10) = \int_0^2 x\,dx = \frac{2^2}{2} = 2$

$$y(t) = \begin{cases} 0, & t < 0 \\ t^2/2, & 0 \le t \le 2 \\ 2, & t > 2 \end{cases}$$

j) The output is zero for $t <$ 0

k) As $t \to \infty$ the output $y(t) =$ 2

l) As t increases above zero, the output initially (decreases, increases).

9. Compute the convolution $y(t) = 2\delta(t-3) * e^{-2t}\cos(10t)u(t)$.

a) $y(t) = 2e^{-2t}\cos(10t)u(t-3)$

b) $y(t) = 2e^{-2t-3}\cos(10t-3)u(t-3)$

c) $y(t) = 2e^{-2(t-3)}\cos(10t-30)u(t-3)$

d) $y(t) = 2e^{-2t+3}\cos(10t+3)u(t+3)$

e) $y(t) = 2e^{-2(t+3)}\cos(10t+30)u(t+3)$

f) $y(t) = 2e^{-2(t-3)}\cos(10t+30)u(t-3)$

Convolution $\Rightarrow$ LTI System

Assume $x(t) = 2\delta(t-3)$ (input)

$h(t) = e^{-2t}\cos(10t)u(t)$

